

Herding as the Contragredient of Conformal Prediction

Peter Cotton

`peter.cotton@microprediction.com`

July 2026

Abstract

Conformal calibration and herding each achieve $O(1/n)$ accuracy from the same finite-sample mechanism: a cumulative discrepancy held bounded by an absolute constant and divided by the sample size. We give the two constructions parallel formal treatments and show that they act on opposite sides of the pairing $\langle h, \nu \rangle = \int h d\nu$: herding steers the empirical measure ν against fixed test functionals, while conformal calibration steers the acceptance functional h — and, in the validity proof, transports the evaluation index — against the fixed empirical measure of scores. The correspondence is stated as a contragredient relation for the permutation action, and it accounts for the linear τ/n dependence penalty of Barber and Pananjady (2026), where blocking arguments yield $\sqrt{\tau/n}$. Since the conformal mechanism (Vovk et al., 2005) predates herding (Welling, 2009), the contragredient is herding.

1 Introduction

Two finite-sample constructions, developed independently, obtain $O(1/n)$ rates where sampling arguments give $O(n^{-1/2})$: the calibration step of conformal prediction, whose mechanism dates to the confidence machines of the late 1990s (Vovk et al., 2005; Lei et al., 2018), and herding (Welling, 2009). This note gives the two mechanisms parallel formal statements (Sections 2 and 3), proves that they are dual actions on the two sides of a single pairing (Section 4), and applies the observation to the analysis of conformal prediction under temporal dependence (Section 5). Interpretive remarks, including the relation to the Bayesian-quadrature reading of Snell and Griffiths (2025), are collected in Section 6.

2 Conformal calibration

Fix a miscoverage level $\alpha \in (0, 1)$ and write $p = 1 - \alpha$. Let $S_1, \dots, S_n$ be calibration scores and S_{n+1} a test score, assumed almost surely distinct, and let $k = \lceil p(n+1) \rceil$, assumed $\leq n$. Split conformal prediction declares coverage when $S_{n+1} \leq q_n$, where $q_n = S_{(k)}$ is the k -th smallest calibration score.

The choice of q_n is a feedback rule on the acceptance test $h_q(s) = \mathbf{1}\{s \leq q\}$.

Lemma 1 (Steering identity). *With $q_n = S_{(k)}$ and the scores distinct,*

$$\sum_{i=1}^n (h_{q_n}(S_i) - p) = k - pn \in [p, 1 + p].$$

In particular the cumulative acceptance-rate discrepancy is bounded by an absolute constant independent of n , and the empirical acceptance rate differs from p by at most $(1 + p)/n$.

Proof. Exactly k of the n calibration scores satisfy $S_i \leq S_{(k)}$, so the sum equals $k - pn$. Since $p(n+1) \leq k < p(n+1) + 1$, we have $p \leq k - pn < 1 + p$. $\square$

Validity rests on a deterministic counting identity concerning the *placement* of the test index.

Proposition 1 (Counting identity). *Let $s_1, \dots, s_{n+1}$ be distinct reals and, for $i \in \{1, \dots, n+1\}$, call placement i covered if s_i is at most the k -th smallest of $\{s_j\}_{j \neq i}$. Then placement i is covered if and only if the rank of s_i among all $n+1$ values is at most k , and consequently exactly k of the $n+1$ placements are covered.*

Proof. Placement i is covered iff at most $k-1$ of the other n values are smaller than s_i , iff the rank of s_i among all $n+1$ values is at most k . The ranks are a permutation of $\{1, \dots, n+1\}$, so exactly k indices qualify. $\square$

Corollary 1 (Marginal validity). *If $(S_1, \dots, S_{n+1})$ is exchangeable with almost surely distinct components, then $\mathbb{P}\{S_{n+1} \leq q_n\} = k/(n+1) \geq 1 - \alpha$.*

Proof. By exchangeability the indicator of the event “placement i is covered” has the same expectation for every i ; average Proposition 1 over i . $\square$

No probabilistic inequality enters before the final step, and the final step is an identity in distribution rather than a concentration bound.

3 Herding

Let $\phi : \mathcal{X} \rightarrow \mathcal{H}$ be a feature map into a Hilbert space and $\mu = \mathbb{E}_P \phi(X)$ a target mean. Herding (Welling, 2009) generates points by the recursion

$$x_t = \arg \max_{x \in \mathcal{X}} \langle w_{t-1}, \phi(x) \rangle, \quad w_t = w_{t-1} + \mu - \phi(x_t), \quad (1)$$

with w_0 given.

Proposition 2 (Herding discrepancy). *For any sequence produced by (1),*

$$\left\| \frac{1}{n} \sum_{t=1}^n \phi(x_t) - \mu \right\| = \frac{\|w_0 - w_n\|}{n}.$$

In particular, if $\sup_t \|w_t\| \leq B$ then the feature-mean discrepancy is at most $2B/n$.

Proof. Telescoping (1) gives $w_n = w_0 + n\mu - \sum_{t \leq n} \phi(x_t)$; rearrange and divide by n . $\square$

Conditions under which the weights remain bounded are given by Welling (2009) in finite dimension and analysed in general via the identification of (1) with the Frank–Wolfe algorithm on the marginal polytope (Bach et al., 2012); Chen et al. (2010) treat the kernel case. Under such conditions herding attains $O(1/n)$ where i.i.d. sampling attains $O(n^{-1/2})$.

4 The duality

Sections 2 and 3 exhibit the same ledger: a cumulative discrepancy — $k - pn$ in Lemma 1, $w_0 - w_n$ in Proposition 2 — bounded by a constant and divided by n . The two mechanisms differ in which argument of the pairing

$$\langle h, \nu \rangle = \int h d\nu$$

they control. Herding fixes the test functionals $h = \langle w, \phi(\cdot) \rangle$ (equivalently, the coordinates of ϕ) and constructs the empirical measure $\nu_n = \frac{1}{n} \sum_t \delta_{x_t}$. Conformal calibration fixes the empirical measure of scores and constructs the test h_{q_n} ; its validity proof additionally moves the evaluation index, as follows.

Let ρ denote the permutation representation of the symmetric group $\mathfrak{S}_{n+1}$ on $\mathbb{R}^{n+1}$, $\rho(\pi)e_i = e_{\pi(i)}$, and let $\rho^*(\pi) = \rho(\pi^{-1})^\top$ be the contragredient representation on the dual space, so that $\rho^*(\pi)e_j^* = e_{\pi(j)}^*$ and $\langle \rho^*(\pi)f, \rho(\pi)v \rangle = \langle f, v \rangle$.

Proposition 3 (Two descriptions of the orbit average). *For $s \in \mathbb{R}^{n+1}$ with distinct entries let $\chi_i(s) = \mathbf{1}\{s_i \leq Q_k(s_{-i})\}$, where $Q_k(s_{-i})$ is the k -th smallest entry of s with coordinate i deleted. Then for every $\pi \in \mathfrak{S}_{n+1}$,*

$$\chi_{n+1}(\rho(\pi)s) = \chi_{\pi^{-1}(n+1)}(s).$$

Consequently the average of χ_{n+1} over rearrangements of the data equals the average of $s \mapsto \chi_i(s)$ over placements i of the evaluation index, and Corollary 1 may be proved on either side of the pairing.

Proof. Q_k is invariant under permutations of its n arguments, and coordinate $n+1$ of $\rho(\pi)s$ is $s_{\pi^{-1}(n+1)}$; substitute. $\square$

Remark 1 (Direction of the duality). Permutation representations are orthogonal, so ρ^* is equivalent to ρ ; the content of the contragredient formulation is not a new representation but the location of the action — on the covector (test) side of the pairing for conformal prediction, on the vector (measure) side for herding. Chronologically the index-side mechanism came first: the conformal construction predates herding by roughly a decade. In that sense it is herding that realises the contragredient of conformal prediction, in the primal space of generated points, with active selection of x_t replacing the transport of an evaluation index.

The correspondence is summarised below.

	Conformal calibration	Herding
controlled side	test h_{q_n} (and evaluation index)	measure ν_n
fixed side	empirical score measure	feature tests ϕ
ledger	$k - pn \in [p, 1 + p)$	$w_0 - w_n, \ w_t\ \leq B$
rate	$(1 + p)/n$	$2B/n$
probabilistic input	exchangeability	none (deterministic target)

5 Dependent data

The dual reading extends to the time-series analysis of Barber and Pananjady (2026), where exchangeability fails and the transport step of Proposition 3 is no longer distribution-preserving. Their argument deletes a buffer of τ coordinates before applying the transport. Two facts price this operation. First, deleting τ of n entries moves the k -th order statistic by at most τ ranks, so the

quantile level shifts by at most $\tau/(n+1)$ — Lemma 1 degrades linearly in the number of damaged slots. Second, the residual failure of transport invariance is charged at its total-variation price, the switch coefficient $\bar{\Psi}_\tau$, bounded by $2\beta(\tau)$ for a stationary β -mixing sequence. The resulting floor,

$$\mathbb{P}(\text{cover}) \geq 1 - \alpha - \min_{\tau} \left\{ \frac{\tau}{n+1} + \bar{\Psi}_\tau \right\},$$

contains no concentration step, and correspondingly no square root: when the same problem is attacked by blocking and empirical-process methods (Oliveira et al., 2024) the rate degrades to $\sqrt{\tau/n}$. The advantage of Barber and Pananjady (2026) over blocking is the advantage of Proposition 2 over Monte Carlo, transported across the pairing.

6 Interpretation and limits

Remark 2 (An image). Herding holds the tests fixed and moves the points; conformal holds the points fixed and moves the test — in a gambling idiom, one moves the balls, the other relabels the slots under the pointer. We record the image once and do not lean on it.

Remark 3 (Quadrature). Snell and Griffiths (2025) recast the conformal computation as Bayesian quadrature over the calibration scores. That is a measure-side reading of the same pairing: quadrature weights on scores on one side, a transported evaluation functional on the other.

Remark 4 (Limits). The duality concerns the finite-sample ledger only. The passage from Proposition 1 to Corollary 1 is exchangeability’s, or, in Section 5, its mixing surrogate’s; no bookkeeping supplies it. Nor does control of thresholds and labels create information: it re-levels coverage while conditional sharpness remains a property of the scores (Cotton). Whether an actively *aimed* index-side controller exists — selecting or reweighting calibration slots to balance the rank count conditionally, as herding aims its particles — is open; localized and weighted conformal methods are the nearest existing steps, and the same information-theoretic ceiling applies to them.

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